

Pranav Srikrishnan

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 GitHub |  LinkedIn |  Portfolio

EDUCATION

K J Somaiya School of Engineering

2024 – Present

B.Tech in Computer Engineering (TY)

CGPA: 8.45

Indian Institute of Technology Madras

BS in Data Science (Foundational Level Cleared); Currently Pursuing Diploma in Data Science

TECHNICAL SKILLS

AI: Generative AI, Gemini API, Prompt Engineering, Search Grounding, Structured Output, MCP Integration

Machine Learning: Data Preprocessing, Hyperparameter Tuning, Feature Engineering, Model Evaluation, Time-Series Modeling, Deep Learning, Transformers, Computer Vision, OCR, Scikit-learn, PyTorch

Programming: Python, C, C++, Java, TypeScript

Data Analysis & Visualization: Pandas, NumPy, EDA, Tableau, PowerBI

Full-Stack Development: React, Node.js, Express, Flask, REST APIs

Databases & Tools: PostgreSQL, MySQL, SQLite, Git, GitHub, Figma, Vercel

PROJECTS

Bhoomi Gat Yaan (BGY) — AI-Powered Mine-Rescue Rover | [Live Demo](#)

Python, PyTorch

- Addressed the need for early hazard and worker detection in underground mines by designing a multimodal system combining sensor time-series, RGB/IR vision, thermal imaging, and localization.
- Compared an LSTM and a Transformer classifier for hazard prediction; the LSTM achieved **94.73% accuracy, 0.9474 F1, and 3.8 ms latency** (Transformer: 93.78%, 0.937, 6.2 ms) and was chosen as the primary model, alongside YOLO + thermal CNN fusion for worker detection.

Meredian — AI-Powered Private Equity Discovery Platform | [Live Demo](#)

Full Stack, MCP, Gemini API

- Addressed the slow, manual process of sourcing private equity targets by building a full-stack platform that converts natural-language investment mandates into structured screening criteria using Gemini with Google Search grounding.
- Delivered automated company screening, shortlist generation, deep-dive investor dossiers, and CSV/PDF reporting, with AI tools integrated through MCP.

Mushroom Classification using Ensemble ML | [Live Demo](#)

Python, Scikit-learn, Pandas

- Built an end-to-end classification pipeline with domain-specific feature engineering, benchmarking multiple models and ensembles; a tuned CatBoost model reached **99.288% accuracy and 0.9985 F1-score**.
- Analyzed model-derived feature importance for interpretability and scored **100/100** in the IIT Madras evaluation.

Flight Price Prediction using Ensemble Machine Learning | [Live Demo](#)

Python, Scikit-learn

- Built an end-to-end price prediction pipeline under IITM with feature engineering and automated preprocessing using Pipeline.
- Benchmarked eight regression models and achieved an **R² score of 0.9703** with a Stacking Regressor.

EXPERIENCE

Smart India Hackathon (SIH) - Internal Round Qualifier | [Live Demo](#)

2026

- Tackled bridge risk prediction by building **Bridge-Sense**, an ML-based Structural Health Monitoring system that uses sensor and contextual data to generate health scores, risk levels, and structural condition classification.
- Selected XGBoost as the primary model, achieving **96.4% accuracy and 0.988 ROC-AUC**, and cleared the internal selection round.

EY Technology Risk - Virtual Job Simulation | [Certificate](#)

Aug 2026

- Completed practical tasks on the fundamentals of technology risk, learning how IT risks are identified and assessed in a client's business.
- Practiced interacting with business stakeholders and asking probing questions to understand processes and uncover risks, then worked in a team to reach and present a conclusion.

Symposium Team Member - SMLRA

Jun 2024 – Oct 2025

- Developed machine learning projects and contributed to seminars on applied machine learning concepts.

ACHIEVEMENTS

National-Level Table Tennis Player

Winner, **International** Carnatic Music & Dance Association

National Award Winner, Akhil Bharatiya Sangh (Mridangam)